Case statement

1. Write an SQL query using CASE to classify each product by its price

* price greater than 1000 → 'Expensive'
* price between 100 and 1000 → 'mid-range'
* price below 100 → 'budget'

Syntax

```
SELECT product_name, price,
    CASE
        WHEN price > 1000
        THEN 'Expensive'
            WHEN price BETWEEN 100 AND 100
            THEN 'Mid-range'
             WHEN price < 100 THEN 'Budget'
    END AS price_category
FROM products;
```

Output:

product_name	price	price_category
laptop	1200.00	Expensive
phone	800.00	Mid-range
keyboard	45.00	Budget
monitor	300.00	Mid-range
mouse	25.00	Budget

2. Write an SQL query using CASE to classify each order

* amount greater than or equal to 1000 → 'High Value'
* amount from 500 to 999.99 → 'Medium Value'
* amount below 500 → 'low Value'

Syntax

```
SELECT customer-name, amount,
    CASE
      WHEN amount >=1000
THEN 'High Value'
    WHEN amount BETWEEN 500 AND 999.9
THEN 'Medium Value'
      WHEN amount < 500
THEN 'Low Value'
      END AS order-value-category
FROM orders;
```

Output.

customer-name	amount	order-value-category
Alice	150.00	Low Value
Bob	560.00	Medium Value
Charlie	999.99	Medium Value
Diana	45.00	low Value
Ethan	1200.00	High Value

3. If the department is 'IT' and salary is greater than 80000 → 'Senior IT'
* If the department is 'HR' and salary is greater than 55000 → 'Experienced HR'
* otherwise → 'staff'

Syntax

```
SELECT emp-name, department, salary,
  CASE
     WHEN department = 'IT' AND salary > 80000
     THEN 'Senior IT'
  WHEN department 'HR' AND salary > 55000
     THEN 'Experienced HR'
     ELSE 'staff'
          END AS position_level
FROM employees;
```

Output.

emp-name	department	salary	position_level
John	IT	85000	Senior IT
Sara	HR	60000	Experienced HR
Mark	IT	75000	staff
Lucy	finance	95000	staff
Tom	HR	55000	staff.

4. * Score greater than or equal to 90 → 'A'
* Score from 80 to 89 → 'B'
* Score from 70 to 79 → 'C'
* Score from 60 to 69 → 'D'
* Score below 60 → 'F'

Syntax

```
SELECT student-name, score,
  CASE
    WHEN score >= 90
THEN 'A'
  WHEN score BETWEEN 80 AND 89
THEN 'B'
  WHEN score BETWEEN 70 AND 79
 THEN 'C'
  WHEN score BETWEEN 60 AND 69
   THEN 'D'
ELSE 'F'
    END AS grade
FROM students;
```

Output:

student-name	score	grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

5* 30 minutes or less → 'Fast'
* 31 - 60 minutes → 'On Time'
* more than 60 minutes → 'Late'

Syntax

```
SELECT delivery-id, delivery-time-minutes,
  CASE
      WHN delivery-time-minutes <=30
    THEN 'Fast'
    WHEN delivery-time-minutes BETWEEN
         31 AND 60
    THEN 'On Time'
    WHEN delivery-time-minutes >60
    THEN 'Late'
  END AS performance
FROM deliveries;
```

Output

delivery-id	delivery-time-minutes	Performance
1	45	On Time
2	80	Late
3	30	Fast
4	65	Late
5	100	Late

6. * Priority 3 → 'High'
* Priority 2 → 'Medium'
* Priority 1 → 'Low'

Syntax

```
SELECT issue_type, priority
  CASE
    WHEN priority = 3
    THEN 'High'
    WHEN priority = 2
    THEN 'Medium'
    WHEN priority = 1
    THEN 'Low'
        END AS priority_label
FROM tickets
```

Output.

issue_type	priority	priority_label
login issue	1	Low
server down	3	High
slow system	2	Medium
Email error	2	Medium
Password reset	1	Low

* 90% or more → 'Excellent'
* 75-89% → 'Good'
* Below 75% → 'Needs Improvement'

Syntax

```
SELECT student_id, (days_present X 100 ÷ total_days) AS
        attendance_percentage,
    CASE
     WHEN (days_present X 100.0 ÷ total_days) >= 90
       THEN 'Excellent'
     WHEN (days_present X 100.0 ÷ total_days) >= 75
       THEN 'Good'
            ELSE 'Needs Improvement'
    END AS attendance_status
FROM attendance;
```

Output

student_id	attendance_percentage	attendance_status
1	90%	Excellent
2	60%	Needs Improvent
3	96%	Excellent
4	50%	Needs Improvement
5	100%	Excellent

8. * stock_qty = 0 → Out of stock
* stock_qty between 1 → low stock
* stock_qty > 5 → In stock

Syntax

```
SELECT product_id, stock_qty,
  CASE
    WHEN stock_qty = 0
THEN 'Out of stock'
      WHEN stock qty BETWEEN
      1 AND 5 THEN 'Low stock'
WHEN stock_qty > 5
      THEN 'In stock'
  END AS stock_status
FROM products_inventory;
```

Output:

product_id	stock_qty	stock_status
1	5	low stock
2	0	out of stock
3	25	In stock
4	10	In stock
5.	3	low stock

9. * 25 or more → 'Large'
* 10-24 → 'Medium'
* Below 10 → 'Small'

Syntax

```
SELECT subject, enrolled_students,
  CASE
    WHEN enrolled_students >= 25 THEN 'Large
  WHEN enrolled_students BETWEEN 10 AND 24
   THEN 'Medium'
  WHEN enrolled_students < 10 THEN 'Small'
   END AS class_size_category
  FROM classes;
```

Output.

subject	enrolled_students	class_size_category
Maths	30	Large
English	25	Large
Science	15	Medium
Art	5	small
History	20	Medium.

10 * If payment_method = 'cash' AND amount >= 200, return 'Eligible for Discount'
* otherwise, return 'Not Eligible'

Syntax

```
SELECT payment-id, payment_method, amount
        CASE
            WHEN payment_method = 'cash' AND
                 amount >= 200
            THEN 'Eligible'
    END AS discount_eligibility
FROM payments;
```

Output.

payment-id	payment_method	amount	discount_eligibility
1	Card	50.00	Not Eligible
2	Cash	200.00	Eligible for Discount
3	Card	150.00	Not Eligible
4	Paypal	75.00	Not Eligible
5	Cash	300.00	Eligible for Discount